MILITARY
SPECIFICATION

MIL-R-13116A(SigC)
15 December 1958
Superseding MIL-R-13116(SigC)
10 November 1953

RECORD, TAPE
(CODE PRACTICE TAPE MC-650( ) AND RECORD SET, TAPE AN/GGA-T1( ))

1. SCOPE

1.1 This specification covers equipment designated Code Practice Tape MC-650( ), Case CS-158( ) and Record Set, Tape AN/GGA-T1( ).

a. Code Practice Tape MC-650( ) consists of fifteen rolls of paper tape, 3/8 inch width, on which doubletrack inked recordings of telegraph code signals suitable for training purposes, have been made and is used with Code Practice Equipments AN/FGC-T1( ) and AN/FGC-T4( ).

b. Record Set, Tape AN/GGA-T1( ) has the same physical conformation as Code Practice Tape MC-650( ), except that coded material on the AN/GGA-T1( ) shall be Allied Communications Procedures (ACP) and is used with AN/FGC-T2( ) and AN/FGC-T3( ).

c. Both types of tape are stored and transported in Case CS-158( ).

1.2 The contractor shall furnish only those components specified in the bid request and contract.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	Tape; paper, gummed (sealing and securing)
LLL-F-291	Fiberboard; corrugated, single face (flexible)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant of Communications, Electronic, and Associated Electrical Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

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STANDARDS

MIL-STD-22	Welded-joint Design
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

SIGNAL CORPS

SC-D-19898
SC-D-19899
SC-D-19900

PUBLICATION

SIGNAL CORPS

List of Military Standards Covering Fasteners.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained by the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification
Consolidated Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W. Washington 6, D.C.)

3. REQUIREMENTS

3.1 Construction.- The equipment covered by this specification shall be constructed as defined herein.

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3.1.1 Case CS-158( ).- Case CS-158( ) shall be constructed in accordance with Signal Corps Drawings SC-D-19898, SC-D-19899 and SC-D-19900. However, if specified performance characteristics cannot be obtained by strict conformance with the drawings, the specified performance characteristics shall govern.

3.2 Parts, materials, and processes.-

3.2.1 Selection.- Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements as specified herein.

3.3 Tropicalization.-

3.3.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.3.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.3.2 Treatment of materials.-

3.3.2.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.3.2.2 Toxicity.- Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.3.2.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.3.2.4 Statement of treatment.- The contractor shall submit, to the contracting

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officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.3.3 Materials not resistant to moisture and fungi. - The following materials are considered not resistant to moisture and fungi:

(a) Felt
(b) Paper

3.4 Paper.-

3.4.1 Quality.- The paper tape shall be of such quality that the ink employed for recording shall be absorbed without smearing or blurring. The paper used shall be made from 100 percent virgin bleached chemical wood pulp, free from groundwood. Dictum Recorder Paper Tape, as made by Link Paper Co., New York, N.Y., or equal, has been found to meet the requirements of this paragraph.

3.4.2 Width.- The width of the paper tape shall be not less than 0.365 inch nor more than 0.385 inch and shall be uniform throughout.

3.4.3 Thickness.- The thickness of the paper tape shall be not less than 0.0033 inch nor more than 0.0042 inch, and shall be uniform throughout.

3.4.4 Color.- The color of the paper tape shall be white.

3.4.5 Rolls.- All rolls of paper tape shall be shear cut and the edges shall be free of lint accumulation.

3.5 Ink.- The ink used for recording the telegraph code signals on the paper tape shall be Higgins American India Ink, waterproof black, #4416, as made by the Higgins Ink Company Inc., 271 Ninth Street, Brooklyn, New York, or equal.

3.5.1 The ink shall be opaque, of the suspension type, shall dry into the paper tape without leaving any ragged edges during recording or after the period of drying, and shall dry to a hard surface not susceptible to smudging when rubbed with the fingers.

3.6 Reels.- The tape reels used for holding the recorded tape shall be constructed of metal and shall be of the reversible type to make rewinding unnecessary when used in photo electric keys such as Keyer KY-127( )/GG.

3.6.1 The reels shall be the commercial 400 foot, 16mm type of motion picture film reel made by Raygram Corp., New York, N.Y., or equal.

3.6.2 All reels shall have a hole for mounting in either direction on a 5/16 inch square shaft.

3.7 Recorded lines.- The coded material on the paper tape shall be an exact duplication of the information contained on the master perforated tapes supplied.

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3.7.1 When paper tape recordings are made the master perforated tapes shall be keyed through a keying head at the rate of 20 words per minute. (The characters of the master perforated tapes supplied have been recroded at the rate of 20 words per mimute. The spacing between characters or groups of characters vary in accordance with words per minute indicated on each individual reel.)

3.7.2 When paper tape recordings are made the tape speed of the recorder shall be adjusted for pulling the tape at exactly 12 feet per minute.

3.7.3 The recorded traces of the telegraph code signals shall be sharp, clean, and square cut, with no evidence of slope to facilitate the use of double track inked recorded tapes in photo-electric keyers such as Keyer KY-127 ( )/GG.

3.7.4 The recordings on any paper tape shall be so made that, when practice transmissions from one recording have been completed, it is only necessary to remove and reverse the reel, rethread, and continue the practice transmissions from the second recording of the same reel without the necessity of rewinding the tape.

3.7.5 The distance from the edge of the paper tape to the nearest edge of the base line shall be not less than 1/16 inch nor more than 5/64. inch.

3.7.6 The recorded line shall be not less than 1/32 inch nor more than 3/64 inch nor more than 1/16 inch.

3.7.7 The signal amplitude from the base line shall be not less than 3/64 inch nor more than 1/16 inch.

3.7.8 There shall be not less than 10 feet of dotted signal recorded on the paper tape, for each track on each reel, preceding and trailing the steady spacing signal to be transmitted by the master perforated tape.

3.7.9 The second recording on each paper tape shall start at a point opposite the end of the first recording on the same surface of the paper tape.

3.7.10 The second recording on each paper tape shall end at a point approximately opposite the start of the first recording on the same surface of the paper tape.

3.7.11 The completed tapes shall be continuous, smooth and uniform and shall contain no splices. The surfaces and edges shall be free from dirt, ink smudges, blots, foreign matter, contain no tears, nicks, perforations, folds, creases or imperfections caused by crushing or crumpling.

3.7.12 The tape of each reel shall be so wound that the start of the first recording, (track "A") is nearest the periphery of the metal tape reel.

3.7.13 The terminal end of the tape of each reel shall be secured to the inner surface of the metal tape reel, near its periphery, with a short piece of brown scotch masking tape.

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3.8 Finish, protective.- Case CS-158( ) shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, etc., and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.7)

3.9 Marking.-

3.9.1 General.- Marking shall conform to Specification MIL-M-13231.

3.9.2 Serial numbers.- Serial numbers under Specification MIL-M-13231 are not required.

3.9.3 The metal tape reel for the MC-650( ) shall be marked with nomenclature "Reel No. ( ) CODEZ Z-( ) Record" or "Reel No. ( ) CODEZ ( )WPM", in letters not less than 3/16 inch in height, in accordance with the information on the master perforated tape reels. The information required for filling in the "( )" shall be obtained from the metal reels on which the corresponding master perforated tapes are wound.

3.9.4 The metal tape reel for the AN/GGA-T1( ) shall be marked with nomenclature "Reel No. ( ) Paris P-( ) Record" or "Reel No. ( ) Paris ( )WPM", in letters not less than 3/16 inch in height, in accordance with the information on the master perforated tape reels. The information required for filling in the "( )" shall be obtained from the metal reels on which the corresponding master perforated tapes are wound.

3.10 Welding.- Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded partt. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever practicable, the welded joint shall conform to Specification MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary.

3.10.1 Cleaning.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.10.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved Inert-gas-shielded arc welding shall be used, whe n practicable, for welding of aluminum or magnesium.

3.11 Rain resistance.- The Case CS-158( ) shall not leak when tested in accordance with 4.6.

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3.12 Interchangeability. - Corresponding components and replaceable parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. (See 4.9) When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular part is to be considered replaceable, the contracting officer shall be consulted.

3.13 Preconditioning. - Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.5. (Also see 4.4(d).

3.14 Preproduction samples. - The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.1(e)(1).)

3.15 Technical literature. - Technical literature shall be furnished as specified in the contract. (See 6.1)

3.16 Workmanship. - The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.1.1 Case CS-158( )
3.3 Moisture and fungus-resistant treatment of paper tapes
3.4 Paper
3.5 Ink
3.6 Reels
3.7 Recorded Lines
3.8 Finish, protective
3.9 Marking
3.10 Welding
3.10.1 Cleaning
3.10.2 Process
3.11 Rain resistance
3.12 Interchangeability

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. - Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and test shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.10)

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(1) Procurement inspection of items before preparation for delivery. (See 4.4)

(2) Procurement inspection of preparation for delivery. (See 4.10 and section 5.)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in the group A and group B inspection specified in tables I and II, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, ballistic or drop and (5) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Procurement inspection of items before preparation for delivery. - The minimum inspection performed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.6) Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A and group B inspection as specified in 4.4.1 through 4.4.2.2.

(b) When Standard MIL-STD-105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when it is considered more practical to select separate samples for group B inspection .

(d) Each unit which will be subjected to group A and group B inspection shall be preconditioned after final assembly/ (See 3.13).

Table I - Group A inspection

Inspection		Rqt. Par.	Insp. Par.	AQL Major	AQL Minor
Visual and Mechanical		3.16	4.8		
Cube CS-158( )		3.1.1	4.8		
moisture and fungus-resistant treatment of paper tape		3.3	4.8	1.0%	4.0%
paper		3.4	4.8		
ink		3.5	4.8		
reels		3.6	4.8		
recorded lines		3.7.3 thru 3.7.13	4.8		
marking		3.9	4.8		
Operational	8	3.7	4.7	1.0%	

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4.4.1 Group A inspection. - This inspection (including sampling) shall conform to table I and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.7) shall be last. (See 6.3)

4.4.2 Group B inspection. - This inspection (including sampling) shall conform to table II and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.4.3 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.4.2.1 Group B sampling plans. - The group B sampling plans, as listed in table II, shall be as follows:

AQL	Inspection level for normal inspection	Inspection level for reduced inspection
4.0%	L-8	L-6

4.4.2.2 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II. - Group B inspection

Inspection	Rqt. Par.	Insp. Par.	AQL
Rain test for Case CS-158( )	3.11	4.6	4.0%
Dimensions	3.12	4.9	6.5%

4.4.3 Disposition of nonconforming product. - When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Bounce preconditioning. - The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L. A. B. Corporation, (Skaneateles, New York) or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the unit shall not be repaired, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.6 Rain test. - Case CS-158( ) shall be subjected to the following tests:

(a) Place the component, in its normal position, in a test chamber maintained at $77^{\circ} \pm 5^{\circ}$F. Allow the simulated rainfall to fall on exposed surfaces of the component. The water shall be at a temperature of $65^{\circ} \pm 5^{\circ}$F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

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(b) With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the component. The water shall fall on exposed surfaces and be blown by the wind against one side of the component. Maintain this condition for 1 hour.

(c) Rotate the component 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

(d) Continue as in 4.6(c) until all four sides of the component have been exposed to the wind and rain, each side for 1 hour.

(e) Open the case. There shall be no visible evidence of leakage.

4.7 Operation test. - Code Practice Tape MC-650( ) and Record Set, Tape AN/GGA-T1( ) shall be subjected, after the completion of the moisture and fungus-resistant treatment, to the following operational tests.

4.7.1 Each reel of recorded tape shall be threaded in a Keyer KY-127( )/GG and run through in both directions in sequence. The recordings shall be checked for exact duplication of the code material on the master perforated tapes and code translation sheets. (The printed text appearing on the code translation sheets is an exact translation of the telegraph code signals as recorded on each master perforated tape supplied and shall be used to prepare duplicate master perforated tapes and to check the accuracy of the information recorded on the paper tape.)

4.7.2 The adjustments of the keyer shall be set at the points where sharp and clear cut signals are produced as the recorded tape is pulled past the scanning aperture. These adjustments shall not be critical when the recorded tape is run through the keyer in both directions, in sequence, and shall apply to all recordings appearing on each tape of each reel of the set under test.

4.8 Visual and mechanical inspection. - Equipment shall be examined for defects listed in table III.

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Table III. - Classification of visual and mechanical defects

Classification	Defect
Major	Smearing or blurring of recording ink.
	Width of recording paper incorrect
	Thickness of recording paper incorrect.
	Color of recording paper incorrect.
	Ink susceptible to smudging when rubbed with fingers.
	Recorded traces of the telegraph code signals not sharp, clean, and square cut, with evidence of slope.
	Color of ink incorrect.
	Distance from the edge of the paper tape to the nearest edge of the base line incorrect.
	Width of the recorded line incorrect.
	Recorded line signal amplitude from the base line incorrect.
	Less than ten feet of dotted signal recorded on the paper. tape, for each track on each reel, preceding and trailing the steady spacing signal to be transmitted by the master perforated tape.
	Recordings on any paper tape made in such a way that rewinding is necessary when one recording on the reel is used following the other.
	Both recordings on each paper tape not on the same surface of the paper tape.
	The second recording of each paper tape starting at a point not opposite the end of the first recording.
	The second recording on each paper tape ending at a point not approximately opposite the start of the first recording.
	The tape of each reel not so wound that the start of the first recording (track A) is nearest the periphery of the metal tape reel.
	Improper treatment for moisture and fungus-resistant properties.
	Improper fit of reels when placed into carrying Case CS-158( ).
	Improper fit of spool when placed on keyer spindle.
	Marking not in conformance with MIL-M-13231.
	Improper operation of fastening devices.
	Other visual and mechanical defects similar to those described above, that are contrary to specified requirements for the equipment.
Minor	The terminal end of tape of each reel not secured to the inner surface of the metal tape reel.

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4.9 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.
(b) Dimensions of cavities, when such dimensions affect insertion of items.
(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.
(d) Size and form of special threads.

4.10 Procurement inspection of preparation for delivery.- The contractor shall perform the inspection specified by 4.10.1 and 4.10.2 to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the inspection specified in 4.10.1 and 4.10.2 to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.6.) When Standard MIL-STD-105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.10.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table IV.

Table IV

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Quantity in unit package not as specified.	2. Packaging material damaged.
3. Incorrect packaging method applied.	3. Conforming wraps are not snug-fitting and contain voids.
4. Cushioning or padding omitted.	
5. Cushioning inadequate for the physical and mechanical protection of the item.	

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Table IV (Contd)

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
6. Stock number omitted, incorrect, or illegible.	4. Any item of marking information other than 6 thru 8 omitted, incorrect, or illegible
7. Nomenclature omitted, incorrect, or illegible.	
8. Marking of quantity of items in package omitted, incorrect or illegible.	
9. Different stock numbered items in the same unit package.	

4.10.2 Packing and marking of exterior containers. - Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table V.

Table V

PACKING, AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Gross weight in excess of specified amount.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Box closure not as specified.	
5. Type, grade, class, and style of the shipping container not as specified.	4. Any item of required marking information other than 10 thru 16 listed under major defects omitted, incorrect, or illegible.
6. Strapping omitted.	
7. Strapping inadequate or incorrectly applied.	

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Table V (Contd)

PACKING, AND MARKING THEREOF

MAJOR	MINOR
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.	
9. Shipping documents or packing list omitted.	
10. Stock number omitted, incorrect, or illegible.	
11. Nomenclature omitted, incorrect, or illegible.	
12. Marking of quantity of items in pack omitted, incorrect, or illegible.	
13. Destination marking omitted, incorrect, or illegible.	
14. Service designation (color marking) omitted.	
15. Specified special marking and labeling, such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
16. Overseas code marking omitted, incorrect, or illegible.	

4.10.2.1 Inspection lot. - A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.10.2.2 Disposition of nonconforming product. - Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.10.2.3 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.1(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

4.6 Rain test
4.7 Operational test

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Record Tape (Code Practice Tape MC-650( )) and Record Set, Tape AN/GGA-T1( ) shall be preserved, packaged and tested in accordance with the procedure specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Code Practice Tape MC-650( ) and Record Set, Tape AN/GGA-T1( ). -

5.1.1.1.1 Stowage in Case CS-158( ). - Stow the record tape within the appropriate compartments of the case. Close the case and secure all fastenings.

5.1.1.1.2 Case CS-158( ) with contents. - Each case with contents shall be packaged method IC-5 as follows: Cushion each case by wrapping in single face flexible corrugated fiberboard conforming to type III of Specification LLL-F-291. Secure wrap with class 3, gummed paper tape, conforming to specification UU-P-111. Place the cushioned case within a close-fitting, regular slotted style, class 2, 275c fiberboard box. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - Record Tape (Code Practice Tape MC-650( )) and Record Set, Tape AN/GGA-T1( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing. -

5.2.1 Level A. - A quantity of Record Tape (Code Practice Tape MC-650( )) and Record Set, Tape AN/GGA-T1( ) shall be packed for shipment within an overseas type or class shipping container as follows: boxes, wood, nailed and lock corners, boxes wood, wirebound, boxes fiberboard wood cleated. Boxes shall conform to the following specifications: PPP-B-621, PPP-B-585, and PPP-B-601. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 200 pounds. Box closure shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Metal strapping. - Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification.

5.2.2 Level B. - Record Tape (Code Practice Tape MC-650( )) and Record Set, Tape AN/GGA-T1( ) packed as specified in 5.2.1 except that the boxes shall be domestic type or class.

5.2.3 Level C. - Record Tape (Code Practice Tape MC-650( )) and Record Set, Tape AN/GGA-T1( ) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

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5.3 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)
(d) The specific paragraphs of section 5 which are applicable to the particular procurement.
(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.14.)
(2) Color chip for finish of Case CS-158( ). (See 6.7.)
(3) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test in accordance with Specification MIL-P-116, which will not be performed as procurement inspection.
d. Five copies of process sheet, using format shown on Drawing SC-C-93073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.
e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required. (See 3.15.)
(i) Facsimiles required.
(j) Submission of the statement of treatment referenced in 3.3, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: MC-650W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

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6.3 Location of operational inspection. - It is desirable that the operational inspection (4.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.4 Inspection. - Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.1 Examination. - Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.2 Testing. - Testing consists of determinations of compliance, using technical means.

6.5 Definitions. -

6.5.1 Inspection terms. - The inspection terms used herein conform to Standard MIL-STD-109.

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.7 Color. - The color chip furnished by the procuring agency will match color chip No. 2430 of superseded Specification TT-C-595, and may be obtained upon request to Commanding Officer, U. S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania, ATTN: SIGSU-J4b. This color chip does not match any color chip of Federal Standard Number 595.

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